coreDAQ

Multi-channel Optical Power Meter

Features

- 4-channel acquisition architecture
- InGaAs / Si detector compatibility
- Linear and logarithmic frontends
- True simultaneous channel sampling
- Deep SDRAM capture buffer
- USB bus-powered operation

Applications

- Optical power monitoring
- Photonic IC characterization
- High-speed capture and logging

Product Overview

coreDAQ is a compact multi-channel optical data-acquisition instrument for lab and production workflows. It provides deterministic, simultaneous sampling across four channels for power monitoring and capture workflows. The platform is USB bus-powered and communicates over USB 2.0 high-speed. Software stack includes coreConsole desktop GUI plus documented Python and C++ APIs for automation. Analog trigger I/O is provided via two analog ports for synchronized measurement integration. Two hardware frontend variants are supported: LINEAR (multi-gain TIA path) and LOG (logarithmic path). Designed for photonics workflows including swept-source measurements, logging, and instrument control integration.

General Specifications

Parameter	Unit	Specification	Notes
Number of channels	-	4 simultaneous channels	Optical channels
Host interface	-	USB 2.0 HS, USB-C	Bus-powered operation
Software	-	coreConsole GUI	Cross-platform desktop
APIs	-	Python API + C++ API	Automation ready
Analog ports	-	2 ports	Trigger / analog integration
Supply	V	5V from USB bus	No external supply required
Operating temperature	°C	10 to 40	Lab environment
Operating humidity	%RH	< 80	Non-condensing
ADC resolution	bit	16	Bipolar conversion path

Sensor / Frontend Specifications

Parameter	Unit	Specification	Notes
Photodiode compatibility	-	InGaAs / Silicon	Device configuration dependent
LINEAR frontend gains	-	8 gain stages	Per-channel gain control
LINEAR model	-	mV = slope*P + intercept	Per gain/channel coefficients
LOG frontend model	-	Parametric log model	LUT fallback if required
Sample rate	ksps	Up to 100 (all channels)	Mode dependent
Capture memory	-	Deep SDRAM buffer	High-density acquisition
Calibration approach	-	Reference PM based	NIST-traceable workflow

Notes:

1) Specifications vary by frontend and detector selection.
2) Calibration is typically referenced at 1550 nm for InGaAs workflows.
3) For silicon detectors, responsivity-curve methods may be used depending on setup.
4) coreConsole controls streaming, capture, calibration, and console command workflows.